

Constructing Many-to-One Vertex-Disjoint Paths in (n, k) -Star Graphs

Tsung-Chi Lin¹, Dyi-Rong Duh^{2,*}, and Hsien-Jone Hsieh³

¹Department of Computer Science and Information Engineering
Nan Kai University of Technology, Caotun, Nantou Hsien 54243, Taiwan
soche@nkut.edu.tw

^{2,3}Department of Computer Science and Information Engineering
National Chi Nan University, Puli, Nantou Hsien 54561, Taiwan
{drduh,s1321902}@ncnu.edu.tw

Abstract

An $S_{n,k}$ refers to a generalized version of an n -star graph (S_n), where $S_{n,n-1}$ and S_n are isomorphic and $S_{n,1}$ is obviously a complete graph of n vertices (K_n). This study constructs a set of many-to-one vertex-disjoint paths in an (n, k) -star graph ($S_{n,k}$), and computes the upper bound of Rabin number of an $S_{n,k}$, where $2 \leq k \leq n-1$. The upper bound of the Rabin number of an $S_{n,k}$ is calculated by constructing a maximal number of many-to-one vertex-disjoint paths from a set of $n-1$ distinct destination vertices to a source vertex in it. This work reveals that if $k=2$, the Rabin number of an $S_{n,k}$ is bounded above by its diameter plus 2, and if $k \geq 3$, bounded by its diameter plus 4 (or 3) for $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and n is even (or otherwise).

1 Introduction

The communication efficiency of an interconnection network greatly depends upon its maximum delay of a message traveling from one node to any other through the network. Thus, routing in an interconnection network is an important issue. Additionally, identifying vertex-disjoint paths in an interconnection network is a parallel routing problem, and has become increasingly important recently in studies of reliability, fault-tolerant ability, maximum parallelism and minimum transmission delay [13]. In general, an interconnection network can be modeled by a graph which can serve as a very powerful mathematical tool for designing and analyzing topological properties of the network. This study attempts to compute the upper bound of the Rabin number of an $S_{n,k}$ by constructing many-to-one vertex-disjoint paths in it.

Let G denote a graph. The vertex set and edge set of G are denoted by $V(G)$ and $E(G)$, respectively. Two vertices u and v are *adjacent* when they are joined by an edge e , where $u, v \in$

$V(G)$ and $e \in E(G)$. All vertices adjacent to a vertex are its *neighbors*. The *distance* from vertex u to vertex v , represented by $d(G; u, v)$, refers to the length of a shortest path from u to v in G . The *diameter* of G , denoted by $d(G)$, is defined as the maximum distance for all pairs of distinct vertices u and v in G [2]. A graph is *connected* when at least one path exists between any two vertices. The *connectivity* of a graph is defined as the minimum number of vertices whose removal renders it disconnected or trivial. Any two paths in a graph are *vertex-disjoint* to each other if they have no common vertex in each path.

Let κ be the connectivity of G . According to Merger's theorem, at least κ vertex-disjoint paths, denoted by $C_\kappa(G; u, v)$, exist between any two distinct vertices u, v in G [2]. The length of a $C_\kappa(G; u, v)$, denoted by $l(C_\kappa(G; u, v))$, is defined as the length of the longest path in $C_\kappa(G; u, v)$ [13]. The $C_\kappa^*(G; u, v)$ is defined as the shortest one among all $C_\kappa(G; u, v)$ s. Let $d_\kappa(G; u, v) = l(C_\kappa^*(G; u, v))$ indicate the *wide distance* from u to v [13]. The *wide diameter* of G , denoted by $d_\kappa(G)$, is defined as the maximum of $d_\kappa(G; u, v)$ s for all pairs of distinct vertices u and v in G .

Rabin number is suggested by Hsu and Lyuu [14], because Rabin first determined this number for the hypercube. Rabin studied the one-to-many vertex-disjoint paths problem in hypercubes and showed that given a source vertex and n destination vertices in an n -dimensional hypercube (Q_n for short), there exist vertex-disjoint paths of length at most $d(Q_n)$ from the source vertex to each of the n destination vertices [22].

Rabin number can be adopted in many kinds of communication patterns, particularly in many-to-one. Let $x, d_1, d_2, \dots, d_\kappa$ be $\kappa+1$ distinct vertices in G . According to Merger's theorem, at least κ vertex-disjoint paths exist between d_i and x for $i=1, \dots, \kappa$ [2]. The Rabin number, denoted by $r_\kappa(G)$, is defined as the minimum number l , such that κ vertex-disjoint paths from d_i to x have length at most l for $i=1, \dots, \kappa$.

An $S_{n,k}$ refers to a generalized version of an S_n , where $S_{n,n-1}$ and S_n are isomorphic [6] and $S_{n,1}$ is obviously a complete graph of n vertices (K_n). An

*Corresponding author

S_n has a regular and hierarchical structure, and has been recognized as an attractive alternative to an n -cube for building massively parallel computers [1], [8]. An S_n has significant advantages, including low degree, small diameter, scalability and symmetry [1]. Additionally, many advantageous properties of an S_n , such as parallel paths [4], [10], spanning trees [23] and a high degree of fault tolerance [11], [12], [18], [24], have been investigated. However, an S_n has a rigorous restriction, i.e., the number of vertices must satisfy $n!$. Thus, additional $(n-1) \times (n-1)!$ vertices are required to expand an S_{n-1} to an S_n . To eliminate this restriction, the new graph $S_{n,k}$ has been presented [6]. An $S_{n,k}$ retains many of the attractive properties of an S_n , but only additional vertices of $(n-1)(n-1)!/(n-k)!$ are utilized to expand an $S_{n-1,k-1}$ to an $S_{n,k}$.

The $S_{n,k}$, where $1 \leq k \leq n-1$, were devised by Chiang and Chen [6]. The structure of an $S_{n,k}$ is formally defined in Section 2. Some basic properties of an $S_{n,k}$, such as diameter [6], connectivity [6], average distance [7], Hamiltonicity [15], spanning connectivity [16], broadcasting [5], embedding [3], optimal spanning tree [19], lower bound of freeback vertex sets [25], have recently been computed or derived. These measurement results demonstrate that an $S_{n,k}$ has excellent topological properties.

Since the connectivity of an $S_{n,k}$ is $n-1$ [6], this work attempts to construct $n-1$ many-to-one vertex-disjoint paths, whose lengths are as short as possible, in an $S_{n,k}$. A set of many-to-one vertex-disjoint paths is a *star-container*. The length of the longest star-container among all constructed star-containers serves as the upper bound of the Rabin number of an $S_{n,k}$.

The remainder of this paper is organized as follows. Section 2 formally describes the topological structure of an $S_{n,k}$. Section 3 presents a many-to-one routing algorithm for any set of n distinct vertices in an $S_{n,k}$ for $2 \leq k \leq n-1$ to construct a star-container from $n-1$ vertices to one vertex. Section 4 determines the upper bound of $r_{n-1}(S_{n,k})$ for $1 \leq k \leq n-1$. Finally, conclusion is drawn in Section 5.

2 Background

An $S_{n,k}$ is an undirected-regular and vertex-symmetrical graph. Parameters n and k , where $1 \leq k \leq n-1$, are applied to determine the topological properties, such as the number of vertices in an $S_{n,k}$ is $n!/(n-k)!$, the connectivity of an $S_{n,k}$ is $n-1$ and the diameter of an $S_{n,k}$ is $2k-1$ if $1 \leq k \leq \lfloor n/2 \rfloor$ or $k + \lfloor (n-1)/2 \rfloor$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ [6]. A vertex ρ in an $S_{n,k}$ can be represented as a label

$p_1 p_2 \dots p_k$. The vertex set $V(S_{n,k})$ is defined as $\{p_1 p_2 \dots p_k \mid p_i \neq p_j, \text{ for } i \neq j \text{ and } p_i, p_j \text{ in } \langle n \rangle\}$. Each element in $\langle n \rangle$ is referred to as a *symbol*. Symbol $\in \langle k \rangle$ ($\in \langle n \rangle - \langle k \rangle$) is an *internal* (*external*) symbol. Moreover, symbol p_i in label $p_1 p_2 \dots p_k$ is as follows: (1) *fixed* if $p_i = i$ (notably, position i is the *desired position* for symbol i); (2) *misplaced* if $p_i \neq i$ and p_i is an internal symbol, and (3) *inner external* if p_i is an external symbol. A symbol not in the label $p_1 p_2 \dots p_k$ is *desired* if it is an internal symbol, and *outer external* if it is an external symbol. For instance, with $\rho = 369482$ in $S_{9,6}$, $p_4 = 4$ is a fixed symbol. Additionally, $p_1 = 3$, $p_2 = 6$ and $p_6 = 2$ are misplaced symbols; symbols 1 and 5 are desired symbols; $p_3 = 9$ and $p_5 = 8$ are inner external symbols, and symbol 7 is an outer external symbol. Vertex ρ is adjacent to ρ_{p_i} and ρ_x by the following adjacent rules.

1. $\rho_{p_i} = p_i p_2 \dots p_{i-1} p_1 p_{i+1} \dots p_k$, for all $2 \leq i \leq k$ (swapping p_1 with p_i), and
2. $\rho_x = x p_2 \dots p_k$, for all x is a desired or outer external symbol (replacing p_1 with x).

The edge connecting ρ and ρ_{p_i} is referred to as i -edge, and ρ_{p_i} is referred to as i -neighbor of ρ . The edge connecting ρ and ρ_x is referred to as 1-edge, and ρ_x is referred to as 1-neighbor of ρ . Figure 1 demonstrates the structure of an $S_{4,2}$.

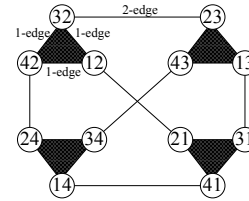


Figure 1. The structure of an $S_{4,2}$.

The degrees of all vertices are $n-1$; each vertex is connected with $(n-k)$ 1-edges, and an i -edge for each i , $2 \leq i \leq k$. Figure 2 illustrates all neighbors of vertex 1234 in $S_{7,4}$.

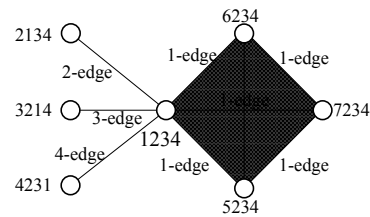


Figure 2. All neighbors of vertex 1234 in an $S_{7,4}$.

For the label of vertex ρ in an $S_{n,k}$, all non-fixed symbols are represented as a cycle representation. The cycle representation is similar to the cycle permutation for an S_n . Restated, when $p_x = y$, $p_y = z$, and $p_z = x$, these non-fixed symbols can be presented as a cycle $(x \ y \ z)$, where x denotes the cycle head, indicating that the desired position of a symbol is occupied by the next symbol in the cycle. Notably, cyclically shifting the symbols in

one cycle does not alter the occupying property of each symbol. Additionally, a label does not adopt the positions from $k+1$ to n , which are called *pseudo-positions*.

For instance, vertex $\rho=369482$ in $S_{9,6}$ can be extended to 369482715, in which outer external symbol 7 is occupied at pseudo-position 7, and desired symbols 1 and 5 are occupied at pseudo-positions 8 and 9, respectively, as indicated by cycles $(1\ 3\ 9\ 5\ 8)(2\ 6)$.

Let $\Phi_i(\rho)=p_i$ denote the symbol at the position i of label ρ . Notably, $\Phi_i(\rho_{p_i})$ and $\Phi_i(\rho_x) \in \langle n \rangle - \{p_1\}$, for all ρ_{p_i} and ρ_x . In the following, some basic properties of $S_{n,k}$ are adopted in this work and can be easily verified according to the adjacent rules.

Fact 1. If vertices u and v are adjacent to each other then $\Phi_1(u) \neq \Phi_1(v)$.

Fact 2. If vertices u and v are two distinct neighbors of vertex ρ then $\Phi_1(u) \neq \Phi_1(v)$.

Lemma 3. Let vertices u and v be any two distinct vertices. If $\Phi_1(u)=\Phi_1(v)$ then $d(S_{n,k}; u, v) \geq 3$.

Proof. Given $\Phi_1(u)=\Phi_1(v)$. Facts 1 and 2 reveal that u and v are not adjacent to each other and have no common neighbor between u and v . Therefore, $d(S_{n,k}; u, v) \geq 3$. $\square$

Lemma 4. Let vertices u and v be two distinct vertices. If both u and v are 1-neighbors of ρ then u and v are also 1-neighbors to each other.

Proof. Without loss of generality, let $\rho=p_1p_2\dots p_k$, $u=xp_2\dots p_k$ and $v=yp_2\dots p_k$, where $x \neq y$ and $x, y \in \langle n \rangle - \{p_1, p_2, \dots, p_k\}$. Obviously, u and v are 1-neighbors to each other, since only the first symbols of u and v are different. $\square$

Lemma 5. In an $S_{n,k}$, a simple cycle has a length at least 6 if it contains one i -edge, $2 \leq i \leq k$.

Proof. Without loss of generality, let $\rho=p_1p_2\dots p_k$, and $\rho_{p_i}=p_1p_2\dots p_{i-1}p_{i+1}\dots p_k$ be two vertices joined by an i -edge e_i in an $S_{n,k}$. Note that $\Phi_1(\rho_{p_i})=p_i$. Suppose that a simple cycle contains ρ and ρ_{p_i} . By the adjacent rules of an $S_{n,k}$, a misplaced symbol p_i should be moved to position 1 and then can be moved to position i . In order to form a path from ρ_{p_i} to ρ excluding the i -edge e_i , a vertex $\rho_{p_i}' \neq \rho_{p_i}$ with $\Phi_1(\rho_{p_i}')=p_i$ should be contained in the path. Thus, the cycle containing ρ and ρ_{p_i} should include ρ_{p_i}' . Since $\Phi_1(\rho_{p_i})=\Phi_1(\rho_{p_i}')=p_i$, $d(S_{n,k}; \rho_{p_i}, \rho_{p_i}') \geq 3$ according to Lemma 3. Therefore, the cycle containing the i -edge e_i has a length at least 6. $\square$

Lemma 6. $1 \leq d(S_{n,k}) - d(S_{n-1,k-1}) \leq 2$.

Proof. Chiang and Chen showed in 1995 that $d(S_{n,k})=2k-1$ if $1 \leq k \leq \lfloor n/2 \rfloor$ and $d(S_{n,k})=k+\lfloor (n-1)/2 \rfloor$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ [6]. Because $1 \leq k \leq \lfloor n/2 \rfloor$ and $\lfloor (n-1)/2 \rfloor + 1 \leq k-1 \leq n-2$ are impossible, three cases

are considered as follows.

Case 1. $(1 \leq k \leq \lfloor n/2 \rfloor$ and $1 \leq k-1 \leq \lfloor (n-1)/2 \rfloor)$:
 $d(S_{n,k}) - d(S_{n-1,k-1}) = 2k-1 - (2(k-1)-1) = 2$.

Case 2. $(\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and $\lfloor (n-1)/2 \rfloor + 1 \leq k-1 \leq \lfloor (n-1)-1 \rfloor)$:

If n is even, $d(S_{n,k}) - d(S_{n-1,k-1}) = (k+n/2-1) - ((k-1)+(n/2-1)) = 1$. If n is odd, $d(S_{n,k}) - d(S_{n-1,k-1}) = (k+(n-1)/2) - ((k-1)+((n-1)/2-1)) = 2$.

Case 3. $(\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and $1 \leq k-1 \leq \lfloor (n-1)/2 \rfloor)$:

If n is even, then $n/2+1 \leq k$ and $k \leq n/2$. It is a contradiction. If n is odd, then $k=(n-1)/2+1$. Hence, $d(S_{n,k}) - d(S_{n-1,k-1}) = (k+(n-1)/2) - (2(k-1)-1) = (n-1)/2 - k + 3 = (n-1)/2 - (n-1)/2 - 1 + 3 = 2$. Thus, the proof is completed. $\square$

Let m and e be the number of misplaced and inner external symbols of vertex ρ , respectively. The cycle representation comprises $m+2e$ symbols in total and implies that $m+2e$ symbols are not in their desired positions. Interestingly, Chiang and Chen showed in 1995 that $d(S_{n,k}; \rho, \varepsilon)$ is $c+m+2e$ if $p_1=1$ or $c+m+2e-2$ if $p_1 \neq 1$, where c denotes the number of cycles. Moreover, they also revealed that $d(S_{n,k})=2k-1$ if $1 \leq k \leq \lfloor n/2 \rfloor$ or $k+\lfloor (n-1)/2 \rfloor$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$.

The vertex $\varepsilon=12\dots k$ is an *identity* vertex whose $p_i=i$ for all $1 \leq i \leq k$. Due to vertex symmetry of an $S_{n,k}$, any vertex can be mapped to the identity vertex ε by a one-to-one mapping function by renaming the symbols. Therefore, without loss of generality, considering the routing problem between any two vertices in an $S_{n,k}$, one of them is always assumed to be the identity vertex ε .

The following first describes the two simple routing rules used to construct a path from ρ to ε in an $S_{n,k}$. A *symbol sequence* is then defined and adopted to construct a shortest path from ρ to ε . The length of the constructed path can be calculated based on the frequency of applied rules.

To construct a shortest routing path π from ρ to ε in an $S_{n,k}$, all misplaced, desired and inner external symbols of ρ should be corrected one by one to transform the label ρ into label ε . The following *shortest-routing rules* are applied to correct symbol s . Each rule can join 1 or 2 edges (vertices) in a path.

R1. If symbol s is at position 1, then

- If s is an internal symbol, then move it to its desired position. (The symbol residing at position s is automatically moved to position 1 when s is corrected.)
- If s is an external symbol, then replace s with a desired symbol. (Symbol s is removed from the label.)

R2. If symbol s is not at position 1, then

- Exchange s with the symbol in position 1, then apply rule R1.

One edge contributes to path π if R1 is applied, and two edges contribute to path π if R2 is applied. For example, let $\rho=369482$ in $S_{9,6}$. If symbol 3 at position 1 is to be corrected, then apply R1 and hop to vertex 963482. If symbol 2 at position 6 is to be corrected, then apply R2 and first hop to vertex 269483 and then reach to vertex 629483. Clearly, all symbols can be corrected in different orders, such that different paths from ρ to ε can be constructed.

Assume that vertex $\rho=p_1p_2\dots p_k$ and its cycle representation is $C_1C_2\dots C_c$. If cycle C_1 contains symbol 1, then regularize symbol p_1 as the head of C_1 , by rotating C_1 . A symbol sequence (sequence for short) is defined as $\sigma=\langle C_1, C_2, \dots, C_c \rangle = \langle s_1, s_2, \dots, s_{m+2e} \rangle$. For example, let $\rho=369482$ in $S_{9,6}$. The cycle representation of ρ is $(1\ 3\ 9\ 5\ 8)(2\ 6)$, and sequence σ is arranged as $\langle C_1, C_2 \rangle = \langle 3, 9, 5, 8, 1, 2, 6 \rangle$, where C_1 is rotated as $(3\ 9\ 5\ 8\ 1)$ and C_2 is $(2\ 6)$.

If path π is constructed in the order σ , then each of $m+2e$ symbols should be corrected by applying R1 or R2. If $p_1=1$, then all c heads are not at position 1 when these heads turn to be corrected and R2 is applied. Conversely, each of the remaining $m+2e-c$ symbols in σ can be corrected by applying R1, since the symbol can be automatically located at position 1 as a result of the correction of its previous symbol in σ . Thus, $2c+(m+2e-c) = c+m+2e$ edges contribute to the constructed path π . If $p_1 \neq 1$, then all c heads except s_1 are not at position 1, and symbol 1 no longer needs to be corrected, since position 1 is its desired position, and $2(c-1)+(m+2e-(c-1)-1) = c+m+2e-2$ edges would contribute to the constructed path π . Let $l(\pi)$ denote the length of path π . Therefore, path π is a shortest path, and $l(\pi)=d(S_{n,k}; \rho, \varepsilon)$.

For example, let $\rho=369482$ in $S_{9,6}$. The cycle representation of ρ is $(3\ 9\ 5\ 8\ 1)(2\ 6)$, in which $c=2$, $m=3$ (symbols 3, 6 and 2) and $e=2$ (symbols 8 and 9). Because $p_1 \neq 1$, $d(S_{n,k}; \rho, \varepsilon) = c+m+2e-2=7$. The sequence σ is arranged as $\langle 3, 9, 5, 8, 1, 2, 6 \rangle$, and path π is constructed according to σ as $369482 \rightarrow_{R1} 963482 \rightarrow_{R1} 563482 \rightarrow_{R1} 863452 \rightarrow_{R1} 163452 \rightarrow_{R2(1)} 263451 \rightarrow_{R2(2)} 623451 \rightarrow_{R1} 123456$, where $\rightarrow_{R1}$, $\rightarrow_{R2(1)}$, and $\rightarrow_{R2(2)}$ signify that rule R1, 1st step of rule R2, and 2nd step of rule R2 are applied, respectively. Since the length of the constructed path is 7, the constructed path is shortest.

3 Many-to-One Routing Algorithm

This section presents the proposed many-to-one

routing algorithm in an $S_{n,k}$ for $2 \leq k \leq n-1$. The algorithm constructs a star-container from a set of $n-1$ distinct source vertices to the destination vertex ε . The algorithm decomposes an $S_{n,k}$ into n mutually disjoint subgraphs. Each subgraph other than the subgraph containing the destination vertex is adopted to construct one path from a source to the destination ε . The length of the longest path among all constructed star- containers serves as an upper bound of Rabin number.

Chiang and Chen [6] indicated that an $S_{n,k}$ can be decomposed into n distinct subgraphs, each subgraph is isomorphic to an $S_{n-1,k-1}$. The vertices in a subgraph have the same symbol at position k . More generally, an $S_{n,k}$ can be decomposed into n distinct subgraphs and the vertices in a subgraph have the same symbol at position i , where $2 \leq i \leq k$. In other words, $V(S_{n,k})$ can be partitioned into n subsets according to the position i of a vertex label. Each subset is denoted by A_j , where $j=1, 2, \dots, n$. A vertex u belongs to A_j if and only if $\Phi_i(u)=j$. Since every vertex in A_j follows the adjacent rules, a subgraph of an $S_{n,k}$ can thus be induced by A_j . This induced subgraph is denoted by $SG_i(j)$. Due to the vertex symmetry of an $S_{n,k}$, the destination vertex can be mapped to the identity vertex ε . Hence, let D be the set of the $n-1$ source vertices, and $\varepsilon \notin D$.

Lemma 7. If $k \geq 3$, there is at least one position t , $t \in \{2, \dots, k\}$, such that an $S_{n,k}$ can be decomposed into n mutually disjoint subgraphs $SG_t(j)$, for $j=1, 2, \dots, n$, and each subgraph contains at most $n-2$ sources of D .

Proof. Suppose that an $S_{n,k}$ is decomposed into n mutually subgraphs according to every position i , $i=2, \dots, k$, and there always exists one subgraph containing all $n-1$ sources of D . This implies that two arbitrary distinct destinations d_x and d_y have $\Phi_i(d_x) = \Phi_i(d_y) = p_i$ for some p_i but $\Phi_i(d_x) \neq \Phi_i(d_y)$. However, after deducted symbols $p_2, \dots, p_k$ from $\langle n \rangle$, only $n-(k-1) \leq n-2$ symbols can be the symbol at position 1. This contradicts to the supposition. Therefore, an $S_{n,k}$ can be decomposed into n mutually disjoint subgraphs, and each subgraph contains at most $n-2$ sources. $\square$

Accordingly, each $SG_t(j)$, $j=1, 2, \dots, n$, contains at most $n-2$ sources. The vertices in $SG_t(j)$ include the symbol j at position t . A vertex $u = p_1p_2\dots p_{t-1}jp_{t+1}\dots p_k$ in $SG_t(j)$ connects to a t -neighbor $v = jp_2\dots p_{t-1}p_1p_{t+1}\dots p_k$ in $SG_t(p_1)$. Subgraph $SG_t(p_1)$ (or $SG_t(j)$) is called the *associated-subgraph* of u (or v) and vertex u (or v) is called an *associated-vertex* of $SG_t(p_1)$ (or $SG_t(j)$).

Lemma 8. Suppose that vertices u_1, u_2 in an $SG_t(j)$ are 1-neighbors each other. u_1 has an i -neighbor u_1'

in $SG_t(j)$ implies u_2 also has an i -neighbor u'_2 in $SG_t(j)$, where $i \in \{2, \dots, t-1, t+1, \dots, k\}$. Both u'_1 and u'_2 are distinct and have the same associated-subgraph.

Proof. Since u_1 and u_2 are 1-neighbors each other, $\Phi_h(u_1) = \Phi_h(u_2)$ for each $2 \leq h \leq k$. Let $u_1 = xp_2 \dots p_{t-1}jp_{t+1} \dots p_k$ and $u_2 = yp_2 \dots p_{t-1}jp_{t+1} \dots p_k$, where $x \neq y$. Without loss of generality, also let $u'_1 = p_ip_2 \dots p_{i-1}xp_{i+1} \dots p_k$ be an i -neighbor of u_1 , where $i \in \{2, \dots, t-1, t+1, \dots, k\}$. Clearly, u_2 has an i -neighbor $u'_2 = p_ip_2 \dots p_{i-1}yp_{i+1} \dots p_k$, and $\Phi_1(u'_1) = \Phi_1(u'_2) = p_i$. Therefore, u'_1 and u'_2 are distinct and the $SG_t(p_i)$ is the associated-subgraph of u'_1 and u'_2 . $\square$

The main idea of the many-to-one routing algorithm can be introduced by a conceptual model. This model points out three problems to be resolved.

Let $D = \{d_1, d_2, \dots, d_{t-1}, d_{t+1}, \dots, d_n\}$ denote the set of source vertices and $SG_t(j)$, $j=1, 2, \dots, n$, be the decomposed subgraphs of an $S_{n,k}$. Also let u_1 be the destination ε and $U = \{u_1, u_2, \dots, u_{t-1}, u_{t+1}, \dots, u_n\}$ denote the vertex set in the $SG_t(t)$, where each vertex u_i in $U - \{u_1\}$ is a neighbor of ε . Each vertex in U joins to its t -neighbor g_i in $SG_t(i)$, where $i=1, 2, \dots, t-1, t+1, \dots, n$. The g_i is the gateway of the $SG_t(i)$. Therefore, $n-1$ vertex-disjoint paths from g_i via u_i to ε can be built as illustrated in Figure 3. Each path has a length at most 2.

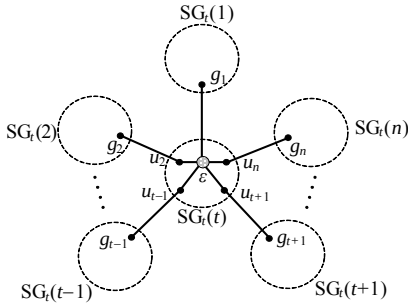


Figure 3. The $n-1$ vertex-disjoint paths from each g_i via u_i to ε .

The subsequence is to assign one subgraph other than $SG_t(t)$ to one source for constructing vertex-disjoint paths between D and ε . Without loss of generality, assuming $SG_t(i)$ is assigned to d_i . If d_i is in $SG_t(i)$ then construct a path from d_i via g_i and u_i to ε directly. This path is represented by $\pi(d_i : g_i, u_i, \varepsilon)$. If d_i is not in $SG_t(i)$ then construct a path from d_i to a vertex o_i in $SG_t(i)$ and then from o_i via g_i and u_i to ε . This path is represented by $\pi(d_i : o_i : g_i, u_i, \varepsilon)$. The subpath from d_i to o_i is called *front-path* of d_i . Suppose that all front-paths are vertex-disjoint, the constructed $n-1$ paths between D and ε are also vertex-disjoint as illustrated in Figure 4. In other words, our goal is to build $n-1$ vertex-disjoint front-paths.

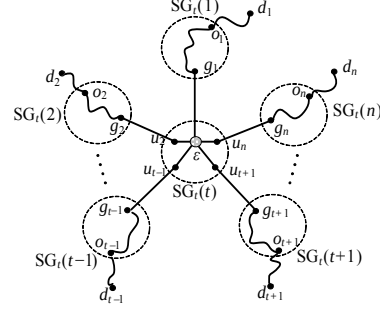


Figure 4. The $n-1$ vertex-disjoint paths between D and ε .

Every gateway is initially annotated as “enabled” and every subgraph $SG_t(i)$, $i \in \langle n \rangle - \{t\}$, is initially annotated as “unmarked”. Subgraph $SG_t(t)$ is annotated as “marked”. If u_x is a source $d_x \in D$, then the edge connecting u_x and ε can be a constructed path from d_x to ε , as shown in Figure 5. In this case, associated-subgraph of d_x , $SG_t(x)$, cannot be assigned to any source, and gateway g_x is annotated as “disabled” and $SG_t(x)$ is annotated as “marked”. Additionally, if a subgraph contains any source, then the subgraph is annotated as “marked”.

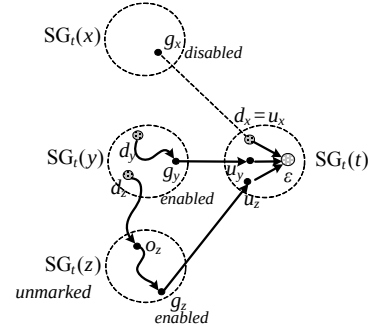


Figure 5. The constructed paths of source $d_x = u_x$, an anchor d_y and source d_z .

If subgraph $SG_t(y)$, $y \in \langle n \rangle - \{t\}$, contains at least one source and gateway g_y is enabled, then subgraph $SG_t(y)$ should be assigned to a source in it. Without loss of generality, $SG_t(y)$ is assigned to d_y . Source d_y is called an *anchor* of $SG_t(y)$, and the constructed path from d_y to g_y is an *anchor-path* of $SG_t(y)$. If a source $d_z \neq d_y$ in $SG_t(y)$, $z \in \langle n \rangle - \{t\}$, then select an unmarked subgraph $SG_t(z) \neq SG_t(y)$ and assign $SG_t(z)$ to d_z for constructing a front-path from d_z to a vertex o_z in $SG_t(z)$. The constructed anchor-path and front-path are shown in Figure 5. If a source is in $SG_t(t)$ but not in U , or a source is in a subgraph whose gateway is disabled, then the path construction is similar to d_z .

According to the discussion above, three problems as follows should be resolved for designing many-to-one routing algorithm:

Problem 1. How to select an anchor for each subgraph with enabled gateway and containing any source?

Problem 2. How to construct a shortest path from a specific vertex to the gateway in a subgraph?

Problem 3. How to assign an unmarked subgraph to a source for constructing a vertex-disjoint front-path whose length as short as possible?

An anchor selection rule is proposed as follows for resolving Problem 1.

Anchor Selection Rule

Let d_y represent the anchor in an $SG_t(y)$, then $d(SG_t(y); d_y, g_y)$ is minimal among every distance from a source to g_y . If more than one source has a minimal distance among all sources to g_y , the source with a marked associated-subgraph has precedence over the others to be the anchor.

Two shortest-routing rules are provided in Section 2, they can be adopted to construct the anchor-path, which is a shortest path from d_y to g_y . Naturally, no other sources are contained in the constructed anchor-path. Moreover, to solve Problem 2, two shortest-routing rules can also be used to construct a shortest path from a specific vertex to the gateway in a subgraph.

Assignment Rules

Assume that an unmarked $SG_t(z)$ is assigned to a source in the $SG_t(t)$, an $SG_t(x)$ or an $SG_t(y)$ for constructing a front-path. Let $a_j, j \geq 1$ be each vertex in the anchor-path, in which $a_1 = d_y$ is the anchor and a_j is adjacent to a_{j+1} . Also let $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$ be the associated-subgraphs of a_1 and a_2 , respectively. Three assignment rules AR1, AR2 and AR3 are provided as follows for resolving Problem 3.

AR1. If $SG_t(z) \neq SG_t(\Phi_1(a_1))$ and $SG_t(z) \neq SG_t(\Phi_1(a_2))$, then assign the $SG_t(z)$ to a source in the $SG_t(y)$ and the source adjacent to a_1 or a_2 has the highest priority.

AR2. If $SG_t(z) = SG_t(\Phi_1(a_1))$ (or $SG_t(z) = SG_t(\Phi_1(a_2))$), then assign the $SG_t(z)$ to a source which is not adjacent to a_1 (or a_2). If more than one source is not adjacent to a_1 (or a_2), the source adjacent to a_2 (or a_1) has precedence over the others.

AR3. The $SG_t(z)$ is assigned to a source in the $SG_t(t)$ or an $SG_t(x)$.

The assigning rules allow a neighbor of a_1 (or a_2) to have the first priority to adopt the unmarked $SG_t(z) \neq SG_t(\Phi_1(a_1))$ (or $SG_t(z) \neq SG_t(\Phi_1(a_2))$). The consideration is to avoid constructing a

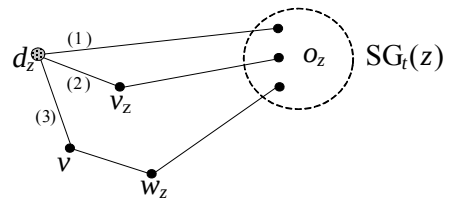
shortest front-path which contains a_1 or a_2 .

Constructing a front-path is described in the following. Suppose that the $SG_t(z)$ is assigned to d_z . If d_z is an associated-vertex of the $SG_t(z)$, the front-path from d_z to o_z only has 1 edge. If d_z is not an associated-vertex of the $SG_t(z)$, the following lemma shows that d_z has a neighbor v_z which is an associated-vertex of the $SG_t(z)$. Notably, v_z can be adopted for constructing a front-path from d_z via v_z to the $SG_t(z)$.

Lemma 9. If d_z is not an associated-vertex of an $SG_t(z)$, then d_z contains a neighbor v_z which is an associated-vertex of the $SG_t(z)$.

Proof. Since d_z is not an associated-vertex of the $SG_t(z)$, $\Phi_1(d_z) \neq z$. By adjacent rules, d_z has a neighbor v_z and $\Phi_1(v_z) = z$. Therefore, v_z is an associated-vertex of the $SG_t(z)$. $\square$

A vertex is *available* if it is not a source and not included in a constructed path. If d_z is not an associated-vertex of the $SG_t(z)$, by Lemma 9, a front-path of length 2 can be constructed from d_z via v_z to o_z when v_z is available. Suppose that v_z is not available, adopt another neighbor v of d_z . Similarly, by Lemma 9, vertex v has a neighbor w_z which is an associated-vertex of the $SG_t(z)$. A front-path of length 3 can be constructed from d_z via v and then w_z to o_z when v and w_z are available. A longer front-path should be constructed if no available vertices v and w_z can be found. Indeed, no front-paths of length longer than 3 are built in this work, which will be shown later. Figure 6 depicts these three paths.



(1) If d_z is an associated - vertex of $SG_t(z)$.
(2), (3) If d_z is not an associated-vertex of $SG_t(z)$.

Figure 6. The different kinds of front-paths from d_z to a vertex o_z .

Let $\pi(d_z : o_z : g_z, u_z, \varepsilon)$ represent a path from d_z to ε containing vertices o_z, g_z and u_z , where symbol “:” (“,”) indicates a subpath (edge). The subpath from d_z to o_z can be constructed as the construction of a front-path. The subpath from o_z to g_z can be constructed by two shortest-routing rules R1 and R2. Obviously, to verify the $n-1$ constructed paths from D to ε are vertex-disjoint, only the paths constructed in each subgraph are needed to be verified.

This algorithm contains two phases. Phase 1 copes with every source u_i in $U - \{u_1\}$ for building a path from u_i to ε . Phase 2 proceeds with five

steps for each subgraph containing some sources. Step 1 finds an anchor and constructs an anchor-path in each $SG_t(y)$. Step 2 constructs a front-path of length 1 from a d_z in an $SG_t(t)$, $SG_t(x)$ or $SG_t(y)$ to o_z in an unmarked $SG_t(z)$. Step 3 builds a front-path of length 2 from a d_z chosen by AR2 in an $SG_t(y)$ to $o_{\phi_1(a)}$ in an unmarked $SG_t(\Phi_1(a))$, where $a = a_1$ or a_2 . Step 4 builds a front-path of length 2 from a d_z in an $SG_t(t)$, $SG_t(x)$ or $SG_t(y)$ to o_z in an unmarked $SG_t(z)$, where $z \neq a_1$ and a_2 . Notably, no front-paths can be established in Step 4, if d_z is adjacent to a_1 or a_2 and no unmarked $SG_t(z)$ s exist. Step 5 first constructs a front-path of length 3 from a d_z is adjacent to a is in an $SG_t(y)$ to $o_{\phi_1(a)}$ in an unmarked $SG_t(\Phi_1(a))$, where $a = a_1$ or a_2 . Second, if any vertex of the path is included in any other path constructed, then these two paths should be reconstructed such that they are vertex-disjoint.

Let $D_i \subset D$ denote the source set in an $SG_t(i)$, $i \in \langle n \rangle$. Also let $D_i^z \subseteq D_i$ contain the associated-vertices of an $SG_t(z)$ for $z \in \langle n \rangle - \{i\}$. The many-to-one routing algorithm is stated as follows.

Algorithm Many-to-One Routing

// Input: A source set D and position t .

// Output: $n-1$ vertex-disjoint paths connecting D and ε .

{

// Initialization:

Set all gateways enabled and all subgraphs unmarked.

Mark $SG_t(t)$.

For each $i \in \langle n \rangle - \{t\}$, mark $SG_t(i)$ if $D_i \neq \emptyset$.

Phase 1. For each source d_x in $SG_t(t)$ and $d_x \in U - \{u_1\}$.

1. Assume that $d_x = u_x$, output the edge connecting u_x and ε as the path $\pi(d_x, \varepsilon)$.
2. Disable g_x and mark $SG_t(x)$.
3. Remove d_x from D_t .

Phase 2. For each $D_i \neq \emptyset$, $i \in \langle n \rangle$.

Step 1. If $i \notin \{t, x\}$

1. Find an anchor of $SG_t(i)$ by anchor selection rule, assuming d_i is the selected.
2. Construct an anchor-path from d_i to g_i by two shortest-routing rules and then augment the path from g_i to ε . The output path is $\pi(d_i : g_i, u_i, \varepsilon)$.
3. Remove d_i from D_i .

Step 2. For each $D_i^z \neq \emptyset$ and $SG_t(z)$ is unmarked.

1. Assign $SG_t(z)$ to a source in D_i^z by the Assignment Rule AR1 if $i \notin \{t, x\}$ and $z \notin \{a_1, a_2\}$, or AR2 if $i \notin \{t, x\}$ and $z \in \{a_1, a_2\}$, or AR3 if $i \in \{t, x\}$, assuming d_z is the source chosen by the Assignment Rule.
2. Construct a front-path of length 1 from d_z to o_z , then augment the path from o_z to g_z by two shortest-routing rules and then from g_z

via u_z to ε . The output path is $\pi(d_z, o_z : g_z, u_z, \varepsilon)$.

3. Mark $SG_t(z)$ and remove d_z from D_i .

Step 3. If $i \in \{t, x\}$, $D_i \neq \emptyset$ and $SG_t(\Phi_1(a_1))$ ($SG_t(\Phi_1(a_2))$) is unmarked.

1. Assign $SG_t(\Phi_1(a_1))$ ($SG_t(\Phi_1(a_2))$) to a source in a $SG_t(i)$ by Assignment Rule AR2, assuming $d_{\phi_1(a_1)}$ ($d_{\phi_1(a_2)}$) is the source chosen by the Assignment Rule.
2. Assume $v_{\phi_1(a_1)}$ ($v_{\phi_1(a_2)}$) is an associated-vertex of $SG_t(\Phi_1(a_1))$ ($SG_t(\Phi_1(a_2))$) and adjacent to the $d_{\phi_1(a_1)}$ ($d_{\phi_1(a_2)}$).
3. Construct front-path of length 2 from $d_{\phi_1(a_1)}$ ($d_{\phi_1(a_2)}$) via $v_{\phi_1(a_1)}$ ($v_{\phi_1(a_2)}$) to $o_{\phi_1(a_1)}$ ($o_{\phi_1(a_2)}$), then augment the path from $o_{\phi_1(a_1)}$ ($o_{\phi_1(a_2)}$) to $g_{\phi_1(a_1)}$ ($g_{\phi_1(a_2)}$) by two shortest-routing rules and then from $g_{\phi_1(a_1)}$ ($g_{\phi_1(a_2)}$) via $u_{\phi_1(a_1)}$ ($u_{\phi_1(a_2)}$) to ε . The output path is $\pi(d_{\phi_1(a_1)}, v_{\phi_1(a_1)}, o_{\phi_1(a_1)} : g_{\phi_1(a_1)}, u_{\phi_1(a_1)}, \varepsilon)$ ($\pi(d_{\phi_1(a_2)}, v_{\phi_1(a_2)}, o_{\phi_1(a_2)} : g_{\phi_1(a_2)}, u_{\phi_1(a_2)}, \varepsilon)$).
4. Mark $SG_t(\Phi_1(a_1))$ ($SG_t(\Phi_1(a_2))$) and remove $d_{\phi_1(a_1)}$ ($d_{\phi_1(a_2)}$) from D_i .

Step 4. When $D_i \neq \emptyset$.

If $i \notin \{t, x\}$, cope with every source d_z in $SG_t(i)$ for building a path from d_z to ε by the following specific priority: (1) d_z is adjacent to a_1 , (2) d_z is adjacent to a_2 , (3) d_z is not adjacent to a_1 and a_2 . Each path is constructed as follows.

1. Assign an unmarked $SG_t(z)$ to a d_z , where $z \notin \{a_1, a_2\}$.
2. Assume v_z is an associated-vertex of $SG_t(z)$ and adjacent to the d_z .
3. Construct a front-path of length 2 from d_z via v_z to o_z , then augment the path from o_z to g_z by two shortest-routing rules and then from g_z via u_z to ε . The output path is $\pi(d_z, v_z, o_z : g_z, u_z, \varepsilon)$.
4. Mark $SG_t(z)$ and remove d_z from D_i .

If $i \in \{t, x\}$, copes with each source d_z in $SG_t(i)$ for building a path from d_z to ε as follows.

1. Assign an unmarked $SG_t(z)$ to a d_z .
2. Assume v_z is an associated-vertex of $SG_t(z)$ and adjacent to the d_z .
3. If d_z in $SG_t(t)$ and $v_z = u_z$ is the neighbor of ε , output the path $\pi(d_z, u_z, \varepsilon)$ directly; otherwise, construct a front-path of length 2 from d_z via v_z to o_z , then augment the path from o_z to g_z by two shortest-routing rules and then from g_z via u_z to ε . The output path is $\pi(d_z, v_z, o_z : g_z, u_z, \varepsilon)$.
4. Mark $SG_t(z)$ and remove d_z from D_i .

Step 5. When $D_i \neq \emptyset$.

Let $SG_t(\Phi_1(a))$ be an unmarked subgraph, where $a = a_1$ or a_2 . Also let v in $SG_t(i)$ be not a source and adjacent to a d_z in $SG_t(i)$.

If at least one of v and a is not an 1-neighbor of d_z , then construct a path from d_z to ε as follows.

1. Assume $v_{\phi_1(a)}$ is an associated-vertex of $SG_t(\Phi_1(a))$ and adjacent to v .
2. Construct a front-path of length 3 from d_z via v and $v_{\phi_1(a)}$ to $o_{\phi_1(a)}$, then augment the path from $o_{\phi_1(a)}$ to $g_{\phi_1(a)}$ by two shortest-routing rules and then from $g_{\phi_1(a)}$ via $u_{\phi_1(a)}$ to ε . The output path is $\pi(d_z, v, v_{\phi_1(a)}, o_{\phi_1(a)} : g_{\phi_1(a)}, u_{\phi_1(a)}, \varepsilon)$.
3. Mark $SG_t(\Phi_1(a))$ and remove d_z from D_i .

When both v and a are 1-neighbors of d_z , then construct a path from d_z to ε as follows.

1. Choose a source d_y is a y -neighbor of d_z , where $y \in \{2, \dots, k\} - \{t\}$, and d_y has a constructed path $\pi(d_y, o_y, \dots, g_y, u_y, \varepsilon)$ from d_y to ε .
2. Assume $y_{\phi_1(a)}$ is an associated-vertex of $SG_t(\Phi_1(a))$ and adjacent to the d_y .
3. Reconstruct a front-path of length 2 from d_y via $y_{\phi_1(a)}$ to $o_{\phi_1(a)}$, then augment the path from $o_{\phi_1(a)}$ to $g_{\phi_1(a)}$ by two shortest-routing rules and then from $g_{\phi_1(a)}$ via $u_{\phi_1(a)}$ to ε . The output path is $\pi(d_y, y_{\phi_1(a)}, o_{\phi_1(a)} : g_{\phi_1(a)}, u_{\phi_1(a)}, \varepsilon)$.
4. If v is available, then construct a path from d_z to ε as follows.

Construct a front-path of length 3 from d_z via v and v_y to an o'_y , where v_y is an associated-vertex of $SG_t(y)$ and adjacent to the v , then augment the path from o'_y to g_y by two shortest-routing rules and then from g_y via u_y to ε . The output path is $\pi(d_z, v, v_y, o'_y : g_y, u_y, \varepsilon)$.

If v is not available, then construct a path from d_z to ε as follows.

- (1). There exists a source d_r is a 1-neighbor of d_z , that d_r has a constructed path $\pi(d_r, v, o_{\phi_1(v)} : g_{\phi_1(v)}, u_{\phi_1(v)}, \varepsilon)$ from d_r to ε .
- (2). Reconstruct a front-path of length 2 from d_r via r_y to an o''_y , where r_y is an associated-vertex of $SG_t(y)$ and adjacent to the d_r , then augment the path from o''_y to g_y by two shortest-routing rules and then from g_y via u_y to ε . The output path is $\pi(d_r, r_y, o''_y : g_y, u_y, \varepsilon)$.
- (3). Construct a front-path of length 2 from d_z via v to $o_{\phi_1(v)}$, then augment the path from $o_{\phi_1(v)}$ to $g_{\phi_1(v)}$ by two shortest-routing rules and then from $g_{\phi_1(v)}$ via $u_{\phi_1(v)}$ to ε . The output path is $\pi(d_z, v, o_{\phi_1(v)} : g_{\phi_1(v)}, u_{\phi_1(v)}, \varepsilon)$.

5. Mark $SG_t(\Phi_1(a))$ and remove d_z from D_i .

}

The constructed many-to-one vertex-disjoint paths are illustrated in Figure 7. The rest of this section is to prove the correctness of this algorithm.

Since an anchor-path is constructed by two shortest-routing rules in Step 1, two lemmas in the

following are obtained.

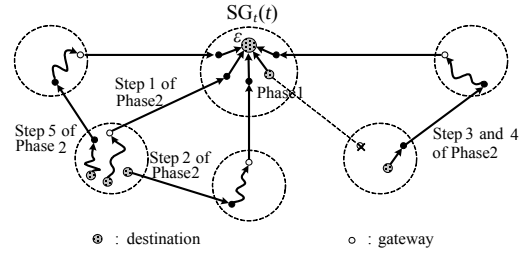


Figure 7. The constructed many-to-one vertex-disjoint paths in an $S_{n,k}$.

Lemma 10. Let a be the vertex a_1 or a_2 , and v_1, v_2 be two distinct vertices in $SG_t(i)$. Assuming v_1 is not lying on the anchor-path. If $d(SG_t(i); a, v_2) > d(SG_t(i); v_1, v_2)$, v_1 is not a source or v_2 is not lying on the anchor-path.

Proof. Suppose that v_1 is a source and v_2 is lying on the anchor-path. According to anchor selection rule, among all sources in $SG_t(i)$, anchor a_1 has the minimum distance to the gateway. Consider $a=a_1$, since v_2 is lying on the anchor-path, $d(SG_t(i); a_1, v_2) > d(SG_t(i); v_1, v_2)$ indicates $d(SG_t(i); a_1, g_i) = d(SG_t(i); a_1, v_2) + d(SG_t(i); v_2, g_i) > d(SG_t(i); v_1, v_2) + d(SG_t(i); v_2, g_i) \geq d(SG_t(i); v_1, g_i)$. However, v_1 is a source, $d(SG_t(i); a_1, g_i) > d(SG_t(i); v_1, g_i)$ yields a contradiction. Consider $a=a_2$, a contradiction is similar to $a=a_1$, because $d(SG_t(i); a_1, g_i) = d(SG_t(i); a_1, a_2) + d(SG_t(i); a_2, v_2) + d(SG_t(i); v_2, g_i) > d(SG_t(i); v_1, v_2) + d(SG_t(i); v_2, g_i) \geq d(SG_t(i); v_1, g_i)$. Therefore, v_1 is not a source or v_2 is not lying on the anchor-path. $\square$

Lemma 11. Let v_1 and v_2 be two vertices in $SG_t(i)$. Assuming v_2 is a source other than anchor, and v_1 is adjacent to v_2 . If $v_1 \neq a_1$ and $v_1 \neq a_2$, v_1 is not lying on the anchor-path.

Proof. Suppose that $v_1 = a_j$, $j \geq 3$, $d(SG_t(i); a_1, v_1) > d(SG_t(i); v_2, v_1)$. This implies $d(SG_t(i); a_1, g_i) > d(SG_t(i); v_2, g_i)$ and yields a contradiction according to anchor selection rule. Therefore, v_1 is not lying on the anchor-path. $\square$

Lemma 12 proves that the paths constructed by Steps 1, 2, 3 and 4 are vertex-disjoint.

Lemma 12. The constructed paths by Steps 1, 2, 3 and 4 are vertex-disjoint.

Proof. Clearly, the paths constructed by Step 1, anchor-paths, are vertex-disjoint. The paths constructed by Steps 2 are also vertex-disjoint since the front-path only joins an edge from a source to its t -neighbor. All front-paths of length 2 are constructed during Steps 3 and 4, and each front-path joints exactly three vertices, denoted by d_z, v_z and o_z . Vertex o_z is a vertex in $SG_t(z)$, and v_z is adjacent to d_z and o_z . Each front-path is vertex-disjoint to the anchor-path, because if $v_z = a_1$ (or a_2), the subgraph $SG_t(z) = SG_t(\Phi_1(a_1))$ (or

$SG_t(\Phi_1(a_2)))$ cannot be assigned to d_z in Step 3. Thus it can be claimed that $v_z \neq a_1$ and $v_z \neq a_2$, and by Lemma 11, v_z is not lying on the anchor-path. Therefore, the constructed front-path is vertex-disjoint to the anchor-path. In addition, after Step 2, an unmarked subgraph $SG_t(z)$ other than $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$, its associated-vertices in $SG_t(i)$ are obviously not a source. Therefore, v_z is not a source. Moreover, because $SG_t(z)$ is assigned to the source d_z , this implies that v_z cannot lie on the other front-paths. Undoubtedly, the constructed front-paths are vertex-disjoint to each other. $\square$

Notably, in Step 4, an unmarked $SG_t(z)$ can be assigned to any source in $SG_t(i)$, because during Step 4, the remaining sources in $SG_t(i)$ are not an associated-vertex of $SG_t(z)$. By Lemma 9, each source has a neighbor which is an associated-vertex of $SG_t(z)$, and by Lemma 11, this neighbor is not lying on the anchor-path. Accordingly, a corollary is obtained in the following.

Corollary 13. In Step 4, an unmarked $SG_t(z)$ can be assigned to any source in $SG_t(i)$.

Corollary 13 indicates that only $SG_t(\Phi_1(a_1))$ or $SG_t(\Phi_1(a_2))$ is unmarked if D_i is not empty after Step 4. In the following, some lemmas are also obtained when Step 4 is completed.

Lemma 14. $D_i = \emptyset$ if $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$ are marked.

Proof. Suppose that $D_i \neq \emptyset$ after Step 4. Corollary 13 indicates that all subgraphs $SG_t(z)$ other than $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$ should be marked. This implies $SG_t(\Phi_1(a_1))$ or $SG_t(\Phi_1(a_2))$ should be unmarked for assigning it to a source of $SG_t(i)$. Therefore, the lemma is proved. $\square$

Lemma 15. If $D_i \neq \emptyset$, $|D_i| = 1$.

Proof. By Corollary 13, $D_i \neq \emptyset$, $SG_t(\Phi_1(a_1))$ or $SG_t(\Phi_1(a_2))$ at least one is unmarked in Step 5. Thus, only $|D_i| = 1$ and $|D_i| = 2$ two cases should be considered. $|D_i| = 2$ is proved as a negative in the following. Because $|D_i| = 2$, both $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$ should be unmarked. Each source in D_i has adjacent to a_1 and a_2 , in contrary, without loss of generality, assuming a source named d_z is the vertex which is not adjacent to a_1 . d_z then has a neighbor $v_{a_1} \neq a_1$ which is available and is an associated-vertex of $SG_t(\Phi_1(a_1))$. A front-path of d_z can thus be constructed from d_z via v_{a_1} to $SG_t(\Phi_1(a_1))$, and $SG_t(\Phi_1(a_1))$ would be marked in Step 3. This yields a contradiction as $SG_t(\Phi_1(a_1))$ is unmarked in Step 5. Therefore, d_z is adjacent to a_1 and a_2 , and $d(SG_t(i); a_1, g_i) = d(SG_t(i); d_z, g_i)$. However, $SG_t(\Phi_1(a_1))$ is unmarked and the associated-subgraph of d_z is marked. According to anchor selection rule, this yields a contradiction as

$d(SG_t(i); a_1, g_i) = d(SG_t(i); d_z, g_i)$. Thus, d_z adjacent to a_1 and a_2 is also a contradiction and $|D_i| = 2$ is a negative. Therefore, $|D_i| = 1$. $\square$

According to Lemma 15, assuming d_z is the only remaining source in D_i if $|D_i| \neq \emptyset$.

Lemma 16. If $SG_t(\Phi_1(a_j))$ is unmarked, d_z is adjacent to a_j , where $j=1$ or 2 .

Proof. Suppose that d_z is not adjacent to a_j . In Step 3, $SG_t(\Phi_1(a_j))$ should be assigned to a source which is not adjacent to a_j . This implies that $SG_t(\Phi_1(a_j))$ has marked after Step 3, and yields a contradiction as $SG_t(\Phi_1(a_j))$ is unmarked after Step 4. $\square$

Lemma 17. If $k=2$, $D_i = \emptyset$.

Proof. Because $SG_t(i)$ and K_{n-1} are isomorphic when $k=2$, all sources in D_i are adjacent to each other, and the anchor-path is joined at most two vertices a_1 and a_2 in $SG_t(i)$. Suppose that $SG_t(\Phi_1(a_1))$ is unmarked, the associated-subgraph of each source is also unmarked according to anchor selection rule. A front-path of length 1 can thus be constructed in Step 2 for each source, and $D_i = \emptyset$ after Step 2. On the other hand, $SG_t(\Phi_1(a_1))$ is marked, the following cases are proved that $SG_t(\Phi_1(a_2))$ also is marked.

Case 1. (g_i is the anchor). All vertices in $SG_t(i)$ except g_i are not lying on the anchor-path, no $SG_t(\Phi_1(a_2))$ should be discussed.

Case 2. (g_i is not the anchor). That $g_i = a_2$, and $SG_t(\Phi_1(a_2)) = SG_t(i)$ is marked in the initialization of the many-to-one routing algorithm.

Since $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$ have been marked in both cases, by Lemma 14, $D_i = \emptyset$. $\square$

Lemma 18. Let source d_y in $SG_t(i)$ and $d_y \neq d_z$. If d_y is not adjacent to a_1 and a_2 , then d_y has a front-path of length 1.

Proof. Suppose that front-path of d_y has a length at least 2. i.e., the associated-subgraph of d_y , denoted by $SG_t(y)$, is not assigned to d_y . This proof shows that there exists no $SG_t(h) \neq SG_t(y)$ which can be assigned to d_y :

Case 1. ($SG_t(h) = SG_t(\Phi_1(a_1))$). It indicates that $SG_t(\Phi_1(a_1))$ is unmarked in Step 1 and is marked after Step 3. Moreover, the associated-subgraph of d_z , denoted by $SG_t(z)$, is marked in Step 1, because d_z has the first priority to apply $SG_t(z)$ rather than other source in Step 2 if $SG_t(z)$ is unmarked. Notably, an unmarked $SG_t(\Phi_1(a_1))$ and a marked $SG_t(z)$ is contradicting to anchor selection rule. Therefore, $SG_t(h) \neq SG_t(\Phi_1(a_1))$.

Case 2. ($SG_t(h) = SG_t(\Phi_1(a_2))$). Subgraph

$SG_t(\Phi_1(a_1))$ should still be unmarked after Step 4. Moreover, the associated-subgraph of d_z , denoted by $SG_t(z)$, is marked in Step 1, because d_z has the first priority to apply $SG_t(z)$ rather than other source in Step 2 if $SG_t(z)$ is unmarked. Notably, an unmarked $SG_t(\Phi_1(a_1))$ and a marked $SG_t(z)$ is contradicting to anchor selection rule. Therefore, $SG_t(h) \neq SG_t(\Phi_1(a_2))$.

Case 3. ($SG_t(h) \neq SG_t(\Phi_1(a_1))$ and $SG_t(h) \neq SG_t(\Phi_1(a_2))$). Source d_z has the first priority to apply $SG_t(h)$ rather than d_y in Step 4. This yields a contradiction as that $SG_t(h)$ is assigned to d_y .

Therefore, d_y has a front-path of length 1. $\square$

Lemma 17 indicates that only $k \geq 3$ should be considered in Step 5. Lemma 15 indicates $|D_i|=1$. Hence, let d_z be the only one source in D_i , and $SG_t(z)$ be the marked associated-subgraph of d_z . Also let a_j denote a_1 or a_2 with unmarked associated-subgraph $SG_t(\Phi_1(a_j))$. By Lemma 16, source d_z is adjacent to a_j . Notably, in $SG_t(i)$, all associated-vertices of $SG_t(\Phi_1(a_j))$ are not a vertex in a constructed front-path, and all associated-vertices of $SG_t(\Phi_1(a_j))$ are not a source, except a_1 if $j=1$.

Lemma 7 indicates that $SG_t(i)$ has at most $n-2$ sources in it, d_z thus has at least one neighbor in $SG_t(i)$, which is not a source and denoted by v . Accordingly, Step 5 is adopted for assigning a subgraph to d_z by some refining methods as follows.

Step 5. $D_i \neq \emptyset$ and all subgraphs are marked except one of $SG_t(\Phi_1(a_1))$ and $SG_t(\Phi_1(a_2))$.

Case 1. (a_j or v at least one is not an 1-neighbor of d_z). Constructing a front-path is as the illustration of Figure 8.

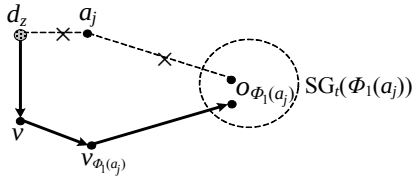


Figure 8. The front-path of the source d_z .

Because at least one of edges a_j-d_z and $v-d_z$ is an i -edge, by Lemma 5, $d(SG_t(i); a_j, v)=2$ and $d(SG_t(i); a_j, v_{\Phi_1(a_j)})=3$. Moreover, $d(SG_t(i); d_z, v)=1$, and $d(SG_t(i); d_z, v_{\Phi_1(a_j)}) \leq 2$, $d(SG_t(i); a_j, v) > d(SG_t(i); d_z, v)$ and $d(SG_t(i); a_j, v_{\Phi_1(a_j)}) > d(SG_t(i); d_z, v_{\Phi_1(a_j)})$. Given d_z is a source, by Lemma 10, the front-path from d_z via v and $v_{\Phi_1(a_j)}$ to $SG_t(\Phi_1(a_j))$ is vertex-disjoint to the anchor-path, and $v_{\Phi_1(a_j)}$ thus is available. Let v' be a neighbor of v . Vertex v' is not adjacent to a_1 and a_2 , because by Lemma 5, a cycle $a_j-d_z-v-v'-a_j$ yields a contradiction as v' is

adjacent to a_1 or a_2 . Suppose that v' is a source, by Lemma 18, v' has a front-path of length 1, and v is available. Therefore, the front-path $\pi(d_z, v, v_{\Phi_1(a_j)}, o_{\Phi_1(a_j)})$ of length 3 can be constructed.

Case 2. (Both a_j and v are 1-neighbors of d_z). This case infers that every i -neighbor of d_z is a source. Each i -neighbor of d_z is not adjacent to a_1 and a_2 , because by Lemma 5, a cycle of length less than 6 yields a contradiction as an i -neighbor is adjacent to a_1 or a_2 . Therefore, by Lemma 18, each i -neighbor has a front-path of length 1. Let source d_y , $y \neq 1$, be an y -neighbor of d_z . The associated-subgraph of d_y (named $SG_t(y)$) is thus assigned to d_y .

Accordingly, this step first proposes a refinement to assign $SG_t(\Phi_1(a_j))$ to d_y , such that $SG_t(y)$ becomes unmarked. The refinement is illustrated in Figure 9, and the front-path of d_y is modified as $\pi(d_y, y_{\Phi_1(a_j)}, o_{\Phi_1(a_j)})$. Vertex $y_{\Phi_1(a_j)}$ adjacent to d_y is an associated-vertex of $SG_t(\Phi_1(a_j))$. By Lemma 5, $d(SG_t(i); a_j, y_{\Phi_1(a_j)})=3$, $d(SG_t(i); a_j, y_{\Phi_1(a_j)}) > d(SG_t(i); d_y, y_{\Phi_1(a_j)})$, and by Lemma 10, $y_{\Phi_1(a_j)}$ is not lying on the anchor-path. $y_{\Phi_1(a_j)}$ thus is available.

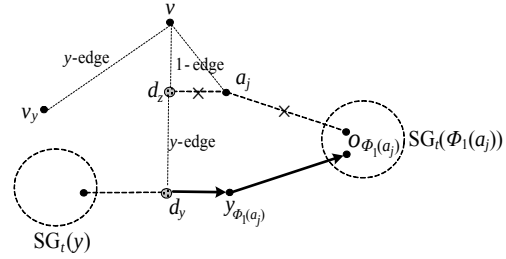


Figure 9. The refined front-paths of source d_y .

Since $SG_t(y)$ is become unmarked, the problem to be concerned with is assigning $SG_t(y)$ to d_z . Given v and d_z are 1-neighbors each other. Source d_z has an y -neighbor d_y , by Lemma 8, v also has an y -neighbor v_y , where $d_y \neq v_y$, both are the associated-vertices of $SG_t(y)$, and they are also illustrated in Figure 9.

In subsequence is to show that v and v_y are not lying on the anchor-path. Given a_j and v are the 1-neighbors of d_z , by Lemma 4, vertices v and a_j are also 1-neighbors each other. Suppose that v is lying on the anchor-path, $v=a_2$ and $a_j=a_1$ since v is not a source. This implies $d(SG_t(i); a_1, g_i) = d(SG_t(i); d_z, g_i)$, but an unmarked $SG_t(\Phi_1(a_1))$ and a marked $SG_t(z)$ is contradicting to anchor selection rule. Therefore, v is not lying on the anchor-path. Moreover, given the anchor-path is the shortest. Suppose that v_y is lying on the anchor-path, by Lemma 5, only the subpath from a_j via v to v_y of length 2 is the shortest. Therefore, v_y is also not lying on the anchor-path since v is

not lying on the anchor-path.

By Lemma 5, v_y is not adjacent to a_1 and a_2 . Suppose that v_y is a source, by Lemma 18 v_y has a front-path of length 1. This indicates that subgraph $SG_t(y)$ would be assigned to v_y rather than d_y in step 2. Therefore, v_y is not a destination and is available.

Assuming v is also available, a front-path $\pi(d_z, v, v_y, o_y')$ of length 3 can thus be constructed as the illustration of Figure 10.

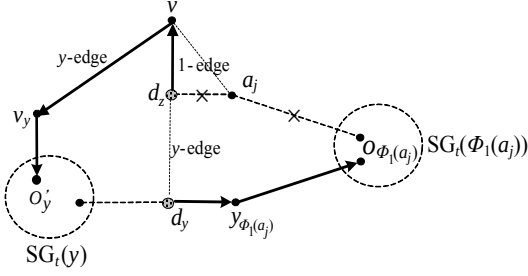


Figure 10. The front-path of source d_z .

On the other hand, assuming v is lying on a constructed front-path from a source d_r via v and $o_{\Phi_1(v)}$ to $SG_t(\Phi_1(v))$, a refinement is adopted to assign $SG_t(y)$ to d_r and $SG_t(\Phi_1(v))$ to d_z . The refined paths are $\pi(d_z, v, o_{\Phi_1(v)})$ and $\pi(d_r, r_y, o_y')$. The illustration of the refinement is shown in Figure 11.

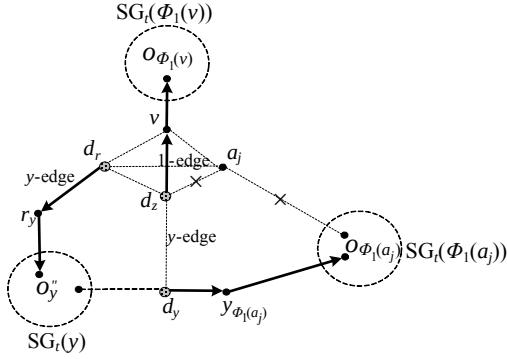


Figure 11. The refined front-paths of source d_r and the front-path of source d_z .

Since $SG_t(\Phi_1(v))$ in Step 4 rather than $SG_t(\Phi_1(a_j))$ in Step 3 is assigned to d_r , this indicates d_r adjacent to a_j . By Lemma 4, d_r, a_j, d_z and v are 1-neighbors each other. Source d_z has an y -neighbor d_y , by Lemma 8, d_r also has an y -neighbor r_y , both d_y and r_y are the associated-vertices of $SG_t(y)$.

Vertex r_y is available, because, (1) By Lemma 5, $d(SG_t(i); a_j, r_y)=2$, and by Lemma 10, $d(SG_t(i); a_j, r_y) > d(SG_t(i); d_r, r_y)$ such that r_y is not lying on the anchor-path. (2) According to Lemma 5, vertex r_y is not adjacent to a_1 and a_2 . Suppose that r_y is a source, by Lemma 18, r_y has a front-path of length 1, this implies that $SG_t(y)$ should be assigned to r_y

rather than d_y in Step 2. Accordingly, the front-path of d_r can be refined as from d_r via r_y to $SG_t(y)$, and the front-path of d_z can be constructed from d_z via v and $o_{\Phi_1(v)}$ to $SG_t(\Phi_1(v))$. According to Lemma 12, $n-1$ vertex-disjoint paths between D and ε are constructed.

4 Rabin Number

An upper bound of $r_{n-1}(S_{n,k})$ is computed in this section. Each path constructed by the many-to-one routing algorithm consists of at most three subpaths. These are the front-path, the shortest routing subpath in a subgraph and the subpath from a gateway to ε . The length of the shortest routing subpath is at most $d(S_{n-1,k-1})$. The length of the subpath from a gateway to ε is at most 2. If $k \geq 3$ (or $k=2$), the front-path constructed in Step 5 (or Step 4) has length of at most 3 (or 2). Totally, the length of each constructed path is at most $d(S_{n-1,k-1})+5$ (or $d(S_{n-1,k-1})+4$). Therefore, $r_{n-1}(S_{n,k}) \leq d(S_{n-1,k-1})+5$ (or $d(S_{n-1,k-1})+4$) if $k \geq 3$ (or $k=2$).

Lin, Duh and Cheng proved that $d_{n-1}(S_{n,n-1})$ is $d(S_{n,n-1})+2$ [20]. Lin and Duh showed that $d_{n-1}(S_{n,k})=d(S_{n,k})+2$ ($d_{n-1}(S_{n,k})=d(S_{n,k})+1$) if $2 \leq k \leq \lfloor n/2 \rfloor$, or $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n < 7$ (if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$) [21]. Based on their results, $r_{n-1}(S_{n,1})$ and the upper bound of $r_{n-1}(S_{n,k})$ for $k \geq 2$ can be obtained in Theorem 19.

Theorem 19. $r_{n-1}(S_{n,k})=d(S_{n,k})$ for $k=1$. If $k=2$, the upper bound of $r_{n-1}(S_{n,k})$ is $d(S_{n,k})+2$, and if $k \geq 3$, the upper bound of $r_{n-1}(S_{n,k})$ is $d(S_{n,k})+4$ ($d(S_{n,k})+3$) when $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and n is even (otherwise).

Proof. If $k=1$, $r_{n-1}(S_{n,1})=d(S_{n,1})=1$ is clearly since $S_{n,1}$ and K_n are isomorphic. If $k=2$, given $d(S_{n-1,1})=1$, $r_{n-1}(S_{n,2}) \leq d(S_{n-1,1}) + 4 = 5 = 2k+1 = d(S_{n,2})+2$. If $k \geq 3$, Lemma 6 indicates $d(S_{n,k})-d(S_{n-1,k-1})=1$ when $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and n is even, or $d(S_{n,k})-d(S_{n-1,k-1})=2$, otherwise. Since $r_{n-1}(S_{n,k}) \leq d(S_{n-1,k-1})+5$, an upper bound of $r_{n-1}(S_{n,k})$ is $d(S_{n,k})+4$ ($d(S_{n,k})+3$) if $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and n is even (otherwise).

5 Conclusion

This work provides a many-to-one routing algorithm for building $n-1$ vertex-disjoint paths from $n-1$ vertices to one distinct vertex for any set of n distinct vertices in an $S_{n,k}$ for $2 \leq k \leq n-1$ such that the length of every constructed path is bounded above by $d(S_{n,k})+2$ for $k=2$, and $d(S_{n,k})+4$ (or $d(S_{n,k})+3$) for $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ and n is even (or otherwise) if $k \geq 3$. The result serves as the upper bound of $r_{n-1}(S_{n,k})$. Moreover, given a graph G

with a connectivity κ , Duh and Chen proved that $r_\kappa(G) \leq \max\{D_\kappa(G), d_\kappa(G)-1\}$ [9], where $D_\kappa(G)$ denotes the κ -1 fault diameter of graph G [17]. Thus, $r_{n-1}(S_{n,k})$ is bounded below by $\max\{D_{n-1}(S_{n,k}), d_{n-1}(S_{n,k})-1\}$. Future work will shorten the constructed paths by the proposed algorithm and compute $D_{n-1}(S_{n,k})$ so that $r_{n-1}(S_{n,k})$ can be determined.

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